

MatchCake: A Python Simulator for Non-Interacting Fermionic Quantum Circuits with Machine Learning Applications

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Summary

We introduce MatchCake, a Python package that simulates matchgate quantum circuits (matchcircuits). Matchgates are a parity-preserving class of nearest-neighbour unitaries whose dynamics are equivalent to those of non-interacting Majorana fermions, making matchcircuits classically simulable in polynomial time while retaining rich quantum structure. Built on PennyLane (Bergholm et al., 2022) and PyTorch (Ansel et al., 2024), MatchCake brings automatic differentiation to this class of circuits, enabling research on classically simulable quantum machine learning models. We demonstrate it on quantum kernel methods for classification (Gince et al., 2024) and on the simulation of physical systems.

Statement of need

Quantum circuits describe the discrete time evolution of a quantum state under a sequence of gates, and are the backbone of quantum algorithms and simulations. Since quantum hardware is still maturing, efficiently simulating and analysing such circuits on classical computers is essential, both for designing quantum algorithms and for mapping the boundary between classical and quantum computational power. In general, the state space of an N -qubit system has dimension $\mathcal{O}(2^N)$, so generic simulators scale exponentially in the number of qubits. Efficient simulation is possible only for restricted circuit families whose states or evolutions admit compact, polynomial-size representations. Matchcircuits are one such family.

A two-qubit matchgate written in the computational basis

$$M(A, W) = \begin{bmatrix} a & 0 & 0 & b \\ 0 & w & x & 0 \\ 0 & y & z & 0 \\ c & 0 & 0 & d \end{bmatrix}, \quad (1)$$

is a parity-preserving nearest-neighbour unitary gate defined by two submatrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, W = \begin{bmatrix} w & x \\ y & z \end{bmatrix}, \quad (2)$$

subject to the constraint $\det(A) = \det(W)$. On a linear register, such matchcircuits evolve as non-interacting Majorana fermions (Dai et al., 2015; Nielsen, 2005). Recent

work (Projansky et al., 2025) connects them to Clifford circuits through fermionic Gaussian operations, framing matchcircuits as a natural generalization that interpolates between stabilizer dynamics and more general fermionic evolutions. Their extensions are also practically relevant: augmenting matchcircuits with ZZ interactions enables efficient simulation of interaction-limited Fermi–Hubbard systems (Mocherla et al., 2023). More broadly, matchgate circuits have found applications spanning quantum thermodynamics (Chapman & Miyake, 2018), quantum machine learning (QML) (Gince et al., 2024; Laar, 2022), and many-body simulation.

State of the field

Simulators differ by the circuit class they handle efficiently. Stim (Gidney, 2021) targets stabilizer (Clifford) circuits with $\mathcal{O}(N^2)$ complexity, while quimb (Gray, 2018) uses tensor-network methods (Biamonte & Bergholm, 2017) such as matrix product states to simulate low-entanglement circuits at a cost of $\mathcal{O}(N\chi^3)$ in the bond dimension χ . PennyLane (Bergholm et al., 2022) is a widely used Python framework for quantum circuit simulation and machine learning, supporting many backends and hardware interfaces, but it lacks efficient native support for matchgate circuits.

Its universal state-vector simulators (SVS) handle arbitrary circuits but scale exponentially in N , and its wrappers around specialized backends such as Stim and quimb do not target matchgates (matchcircuits (Valiant, 2001)). Users are therefore forced onto exponentially scaling simulators, even though matchcircuits admit polynomial-time $\mathcal{O}(N^3)$ classical simulation (Bravyi, 2008; Brod, 2016; Hebenstreit et al., 2020). Unlike tensor-network methods, whose efficiency relies on limited entanglement, matchcircuits derive their tractability from an underlying free-fermionic (Gaussian) structure and so support arbitrarily large entanglement. And unlike the discrete Clifford gate set, matchgates (Jozsa & Miyake, 2008) form a continuously parametrized family of two-qubit gates corresponding to free-fermionic operations. MatchCake fills the remaining tooling gap with a dedicated matchcircuit simulator integrated with PennyLane and PyTorch, whose automatic differentiation supports learning-based models that exploit matchgate dynamics.

Software design

Using MatchCake, the user provides a matchcircuit C (a list of matchgates), a product initial state $|\psi_{\text{initial}}\rangle$, and an observable

$$\mathcal{F} = \sum_j \beta_j F_j \quad (3)$$

with real coefficients β_j , whose terms are computational-basis projectors or Pauli words,

$$F_j \in \{|y\rangle\langle y| : y \in \{0, 1\}^N\} \cup \{P\}, \quad (4)$$

where $|y\rangle$ is a computational-basis state and P any Pauli word on the N -qubit register. MatchCake then computes, in time polynomial in N , the expectation value

$$\text{Output} = \langle \psi_{\text{initial}} | C^\dagger \mathcal{F} C | \psi_{\text{initial}} \rangle. \quad (5)$$

As shown in Figure 1, $\mathcal{F}$ is evaluated term by term. The circuit C is mapped to its transition matrix R_C (Brod, 2016), and each term is Majorana-decomposed through the Jordan–Wigner

64 transformation (Jordan & Wigner, 1928; Nielsen, 2005). When $|\psi_{\text{initial}}\rangle$ is a computational-
 65 basis state and F_j is diagonal in that basis, the per-term expectation ε_j is read from a lookup
 66 table of Wick's contractions (Brod, 2016; Wick, 1950); otherwise it is obtained from the
 67 covariance matrix of the circuit-evolved state. Both routes reduce to Pfaffian computation,
 68 and the per-term values are accumulated as $\sum_j \beta_j \varepsilon_j$.

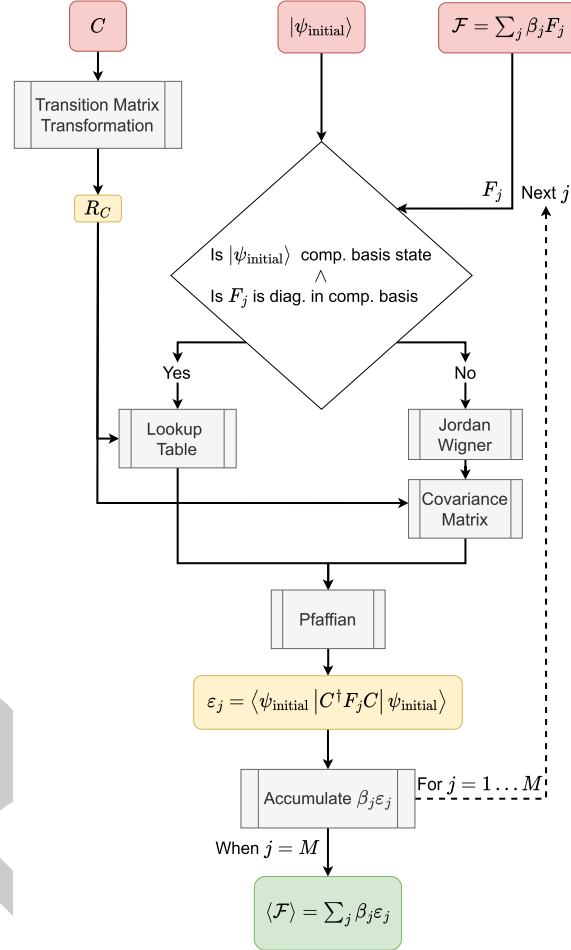


Figure 1: The MatchCake algorithm, which evaluates the expectation value $\langle \mathcal{F} \rangle = \sum_j \beta_j \varepsilon_j$ of the observable $\mathcal{F}$, one Pauli string F_j at a time. When the initial state $|\psi_{\text{initial}}\rangle$ is a computational-basis state and F_j is diagonal in the computational basis (Yes), the per-term expectation ε_j is read from a lookup table; otherwise (No), it is obtained from the covariance matrix of the circuit-evolved state. Both routes share a common Pfaffian evaluation.

69 Minimal Example - Quantum Circuit Simulation

70 The following example builds a matchcircuit and computes the expectation value of a
 71 computational-basis projector after applying the circuit to an initial state.

```

import matchcake as mc
import pennylane as qml
import numpy as np
from pennylane.ops.qubit.observables import BasisStateProjector

# Create a Non-Interacting Fermionic Device with 4 qubits/wires
nif_device = mc.NonInteractingFermionicDevice(wires=4)

```

```
# Define the initial state as the all-zero computational basis state: |0000>
initial_state = np.zeros(len(nif_device.wires), dtype=int)

# Define a quantum circuit
def circuit(params, wires, initial_state):
    # Prepare the initial state
    qml.BasisState(initial_state, wires=wires)
    for i, even_wire in enumerate(wires[:-1:2]):
        idx = list(wires).index(even_wire)
        curr_wires = [wires[idx], wires[idx + 1]]
        # Apply the matchgate M(Rx(params), Rx(params))
        mc.operations.CompRxRx(params, wires=curr_wires)
        # Apply the matchgate M(Ry(params), Ry(params))
        mc.operations.CompRyRy(params, wires=curr_wires)
        # Apply the matchgate M(Rz(params), Rz(params))
        mc.operations.CompRzRz(params, wires=curr_wires)
    for i, odd_wire in enumerate(wires[1:-1:2]):
        idx = list(wires).index(odd_wire)
        mc.operations.fSWAP(wires=[wires[idx], wires[idx + 1]])
    projector: BasisStateProjector = qml.Projector(initial_state, wires=wires)
    return qml.expval(projector)

# Create a QNode
nif_qnode = qml.QNode(circuit, nif_device)
qml.draw_mpl(nif_qnode)(
    params=np.array([0.1, 0.2]),
    wires=nif_device.wires,
    initial_state=initial_state
)

# Evaluate the QNode
expval = nif_qnode(
    params=np.array([0.1, 0.2]),
    wires=nif_device.wires,
    initial_state=initial_state
)
print(f"Expectation value: {expval:.4f}")
```

72 Output: Expectation value: 0.9901

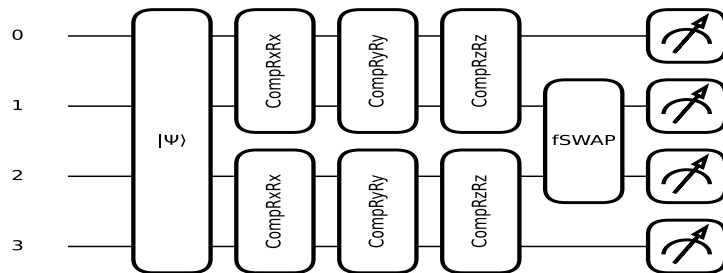


Figure 2: Quantum circuit generated by the circuit function.

73 Minimal Example - QML

74 The next example builds quantum kernel classifiers with the `FermionicPQCKernel` and
 75 `LinearNIFKernel` classes and evaluates them on the iris dataset (Fisher, 1936) using 20-fold
 76 cross-validation. `MatchCake.ml` builds on Scikit-learn (Buitinck et al., 2013; Pedregosa et al.,
 77 2011), giving a familiar interface for machine learning practitioners.

```
import matplotlib as mpl
from matchcake.ml import CrossValidation, CrossValidationVisualizer
from matchcake.ml.kernels import FermionicPQCKernel
from matchcake.ml.kernels.linear_nif_kernel import LinearNIFKernel
from matchcake.ml.visualisation import ClassificationVisualizer
from matplotlib import pyplot as plt
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import MinMaxScaler
from sklearn.svm import SVC

# Load the iris dataset
dataset = datasets.load_iris(as_frame=True)
X, y = dataset.data, dataset.target

# Define estimators using different kernels for comparison
n_qubits = 8
estimators = {
    "Fermionic PQC": Pipeline(
        [
            ("scaler", MinMaxScaler(feature_range=(0, 1))),
            # Example of kernel without alignment, which is the default behaviour
            ("kernel", FermionicPQCKernel(n_qubits=n_qubits, rotations="X,Z")),
            ("classifier", SVC(kernel="precomputed")),
        ]
    ),
    "Fermionic Linear": Pipeline(
        [
            ("scaler", MinMaxScaler(feature_range=(0, 1))),
            # Example of kernel with alignment, which improves the performance
            ("kernel", LinearNIFKernel(n_qubits=n_qubits, alignment=True)),
            ("classifier", SVC(kernel="precomputed")),
        ]
    ),
}

# Run cross-validation
cvo = CrossValidation(estimators, X, y).run()

# Create subplots for visualization
fig, axes = plt.subplots(nrows=1, ncols=2, figsize=(12, 5))
# Visualize the classification boundaries for a selected model and fold
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0
)
viz = ClassificationVisualizer(x=X, n_pts=25_000)
viz.plot_2d_decision_boundaries(
    model=estimators["Fermionic Linear"].fit(x_train, y_train),
```

```

title="",
legend_labels=dataset.target_names,
y=y,
fig=fig,
ax=axes[0],
legend_loc="lower right",
)
# Visualize cross-validation results with violin plots
cv_viz = CrossValidationVisualizer(cvo)
cv_viz.plot(
    ax=axes[1],
    score_name="Accuracy",
    estimator_name_key="Kernel",
    score_name_map={"train_score": "Train", "test_score": "Test"},
    palette="pastel",
)

```

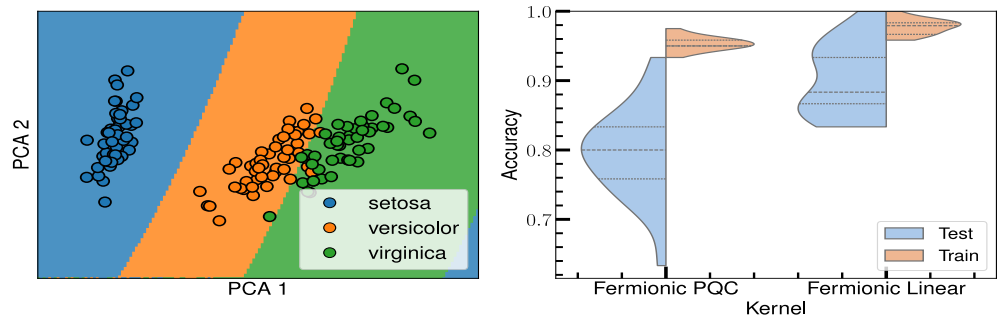


Figure 3: Output of the machine learning example, using 8-qubit kernels and 20-fold cross-validation. Left: decision boundaries for the Fermionic Linear classifier on the iris dataset. Right: violin plots of training and testing accuracies across the cross-validation folds for the Fermionic PQC and Fermionic Linear classifiers.

Research impact statement

MatchCake was central to “Fermionic Machine Learning” (Gince et al., 2024), enabling the benchmarking of kernel-based free-fermionic classifiers at qubit counts infeasible with SVS.

That work underscores the need for classically simulable baselines when benchmarking QML, to test whether claimed quantum resources are actually necessary before asserting a quantum advantage. It also highlights a key strength of free-fermionic models: the provable absence of barren plateaus (Díaz et al., 2023), critical for efficient training. MatchCake will therefore be an invaluable tool for the quantum simulation and QML communities, supporting fundamental research and the empirical validation of quantum advantage claims.

AI usage disclosure

The author used AI-assisted tools to improve the grammar, style, and clarity of the manuscript, and to help draft limited parts of the source-code documentation. No scientific content, results, or conclusions were generated by AI tools. The author has reviewed, edited, and validated all AI-assisted content and assumes full responsibility for the final version of the manuscript and the software.

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